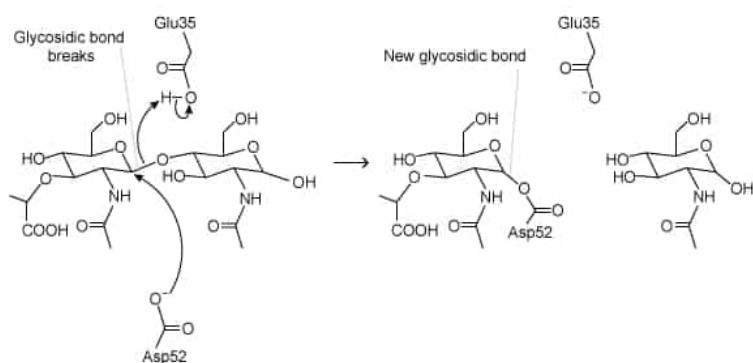


The first step of an enzyme-catalyzed cleavage of a glycosidic bond is shown below.



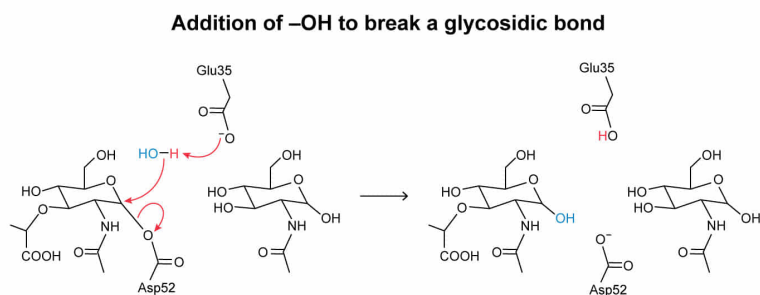
In the second step, Glu35 creates a good nucleophile that breaks the glycosidic bond formed by Asp52 in the first step. Given that the local environment of the enzyme significantly increases the pK_a of the Glu35 side chain, which of the following molecules does Glu35 most likely deprotonate to produce the nucleophile?

- ☐ A. NH_3 [15%]
- ☐ B. H_3O^+ [29%]
- ✓ ☒ C. H_2O [39%]
- ☐ D. OH^- [16%]

Correct

38% answered correctly

Explanation:



Glu35 deprotonates water to form OH^- , a good nucleophile.

A **glycosidic bond** is the α - or β -linkage between the **hemiacetal** or **hemiketal** group of a sugar molecule and a **hydroxyl ($-\text{OH}$) group** of another molecule. Glycosidic bonds are **broken via hydrolysis** reactions (addition of H_2O), frequently with the aid of an enzyme, resulting in two smaller components.

The question shows that in the cleavage of the glycosidic bond between the two given sugar molecules, glutamic acid at position 35 (Glu35) donates a proton to one of the sugars in the disaccharide as aspartate at position 52 (Asp52) acts as a **nucleophile** to attack the **anomeric carbon** of the other sugar. This yields an intermediate in which a sugar is glycosidically bonded to Asp52.

To complete the reaction, Glu35 must return to its protonated form and a hydroxyl group must act as a nucleophile to break the bond to Asp52. This can be accomplished if Glu35 deprotonates a water (H_2O) molecule. The increased $\text{p}K_a$ of Glu35 in its local environment makes this possible. This deprotonation creates a good nucleophile (OH^-) that can then attack the anomeric carbon while simultaneously restoring Glu35 to its protonated form. OH^- then attacks and Asp52 acts as the leaving group.

(Choice A) NH_3 is not involved in the hydrolysis of glycosidic linkages.

(Choice B) Glu35 can deprotonate H_3O^+ , but this yields H_2O , which is *not* a good **nucleophile**. Because cleavage of the glycosidic bond requires a good nucleophile, deprotonation of H_3O^+ is not a good candidate.

(Choice D) Glu35 cannot deprotonate hydroxide (OH^-) in the

second step because Glu35 and OH^- are both negatively charged, so neither can act as an electrophile. Deprotonation of OH^- would yield O^{2-} , which is extremely unstable and typically does not form.

Educational objective:

A glycosidic bond is the α - or β -linkage between a sugar and an –OH of another molecule. Hydrolysis of a glycosidic bond is cleavage of the linkage by addition of H_2O , breaking the molecule into two smaller units.

Time spent:0 sec

Subject:
Organic Chemistry

QID:402733

System:
Functional Groups and Their Reactions